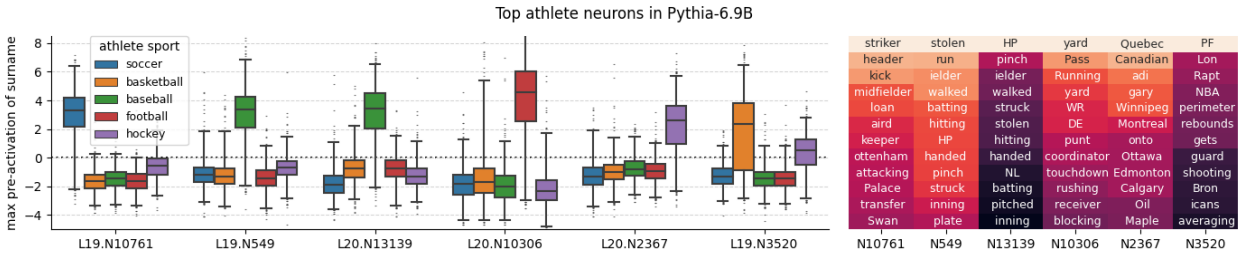
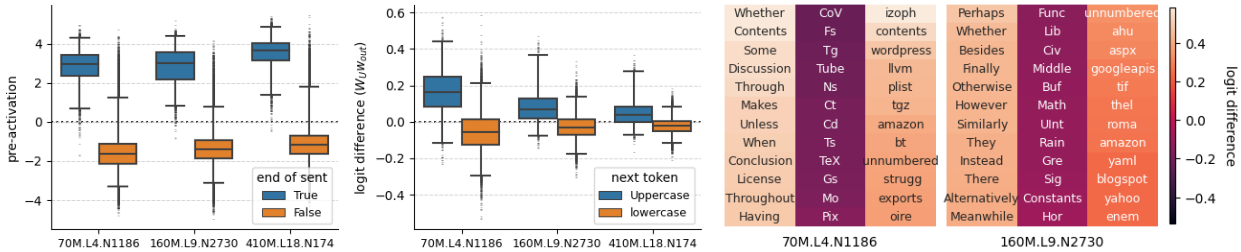
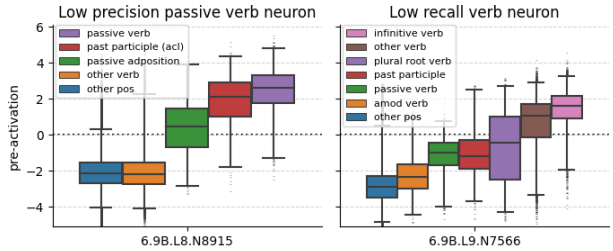


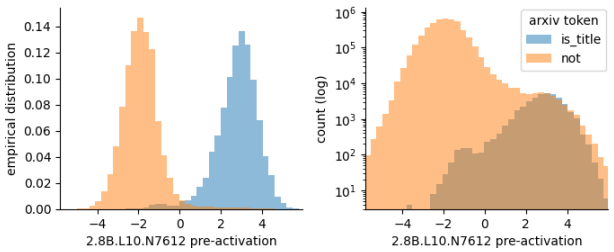
(a) Neurons which activate for the names of specific types of athletes (left) turn out be more general sport neurons when analyzing the top per-token average activations (right).



(b) End of sentence neurons: analyzing the effect on the output vocabulary can corroborate and refine neuron interpretations. For token heatmaps (right), the columns correspond to capital tokens with max increased logits, capital tokens with max decrease, and lowercase tokens with max increase, respectively.



(c) Precision and recall metrics can guide further analysis.



(d) Distributional statistics can hide rare features.

Figure 5: Examples of analyses used to refine and support neuron interpretations.

a person's gender and whether they are alive or not (Figure 20). It is likely more natural for a model to represent the `is_living_female_soccer_player` feature as simply a composition of three different neuron aligned-features `is_living`, `is_female`, and `plays_soccer`, which would nevertheless have low 1-sparse accuracy but high 3-sparse accuracy. Of course, this also assumes that these neurons are all in the same layer. However, models can, and almost certainly do, leverage superposition and composition across multiple layers, a subtlety which we blissfully ignore and leave for future work.

For the remaining discussion, we consider individual neurons identified by 1-sparse probes—how do we interpret such neurons? The most basic analysis is to simply inspect the maximum activating dataset examples [73], though this is subject to interpretability illusions [72]. A slightly more robust analysis involves computing the average activation for each token in the vocabulary. Doing so for our athlete neurons in Figure 5(a) reveals these neurons are perhaps better thought of as generalized sport neurons which activate for all tokens generally having to do with a particular sport, including the names of athletes (with the notable exception of the hockey neuron which appears to be a Canadian neuron).

In addition to analyzing a neuron via the input, one can also gain insight by analyzing the output; in particular, one can compute a neuron's effect on the output logits by simply considering the product of the unembedding matrix and the neuron output weight [50]. Doing so for neurons which always activate on punctuation ending a sentence, we find that the output probability of most uppercase tokens gets increased while the probability of most lowercase tokens gets decreased (Figure 5(b)). Inspecting specific examples is often constructive; for instance, the highest increasing lowercase tokens are words like "amazon" or "wordpress," indicating such neurons are also useful in constructing URLs.

For uppercase tokens, those with the highest increase in probability correspond to words like "Finally" or "However" while the most decreased tokens are those representing atomic elements, proper nouns, or CamelCase strings.

Most of the neurons discussed thus far have had excellent overall classification performance; what happens if there does not exist such a crisp fit? As discussed in Section 3.3, classification performance can be further decomposed into precision (sensitive to false positives) and recall (sensitive false negatives). In our context, high recall and low precision with respect to a specific feature potentially suggests that a neuron represents a less granular feature than the feature manifest in the probe dataset [10]; low recall and high precision suggest a neuron represents a more specific feature. As an example, Figure 5(c) depicts the activations for a high-recall-low-precision neuron identified when probing for `is_passive_verb` and a low-recall-high-precision neuron identified when probing for `is_verb` in Pythia 6.9B. When we analyzed these neurons in more detail, we find that in addition to activating on all occurrences of passive verbs, `L8.N8915` also activates on adjacent adposition tokens, and past participles when in an adnominal clause. Almost symmetrically, `L9.N7566` activates on most verbs, especially infinitive verbs, but not passive verbs, adjectival modifiers, plural root verbs, or past participles in certain dependency roles. These examples illustrate the more general point that it should not be assumed that a model will learn to represent features in an ontology convenient or familiar to humans.

Of course, precision and recall are only defined with respect to the labels of a probing dataset, which may themselves contain imbalances or spurious correlations that confound the results. Perhaps most common, especially when constructing a dataset from scratch, is the problem of asymmetric sampling and rare features. Consider a probing dataset for an `is_arxiv_paper_title` feature (Figure 5(d)). Without a more specific hypothesis on the relevant negative examples, one is likely to simply sample random non-title tokens (or at least weight all non-title examples the same). However, when the space of negatives is large (e.g., all non-title tokens), at a distribution level, all rare features get drowned out. Hence, when looking at summary or distributional statistics, the results can look impressive (Figure 5(d); the neuron had F1 > 0.95 for `is_title`), but when looking at raw counts (Figure 5(d) right) one observes that the `is_title` feature actually explains less than half of the activations. Though, upon inspection, we find that the rest of the activations are for *section* titles!

6 Discussion

6.1 Strengths and Weaknesses of Sparse Probing

The primary use case of sparse probing is to quickly and precisely localize neurons relevant to a specific feature or concept, while naturally accounting for superposition and composition. The speed and precision of recovery is in contrast to gradient-based [38] and causal-intervention based methods [69], which are too slow or coarse-grained to be used for features requiring precise localization within large models. The feature specificity contrasts with sparse autoencoding based methods which aim to recover *all* features stored in the model, but in an unsupervised manner and without attributing semantic meaning to each feature [53]. Having probes with optimality guarantees further addresses the pitfall raised by [13] regarding the conflation of classification quality and ranking quality when analyzing individual neurons with probes. Moreover, sparse probes are designed to be of minimum capacity, mitigating the concern that the probe is powerful enough to construct its own representation to learn the task [14]. While probing requires a supervised dataset, once constructed, it can be reused for interpreting any model (modulo features regarding tokenization), enabling investigation into the universality of learned circuits [5, 46] and the natural abstractions hypothesis [80]. By design, sparse probing is particularly well suited for studying superposition, and can be used to automatically test the effect of architectural changes on the frequency of polysemanticty and superposition, as opposed to relying on human evaluations as in [10].

However, sparse probing also inherits many of the weaknesses of the general probing paradigm [14, 28, 15, 29, 30]. In short, the results from a probing experiment do not allow for drawing especially strong conclusions without more detailed secondary analysis on the identified neurons. Probing offers limited insight into causation, and is highly sensitive to implementation details, anomalies, misspecifications, and spurious correlations within the probing dataset. For interpretability specifically, sparse probes cannot detect features built up over multiple layers or easily distinguish between features in superposition versus features represented as the union of multiple independent and more granular features [12]. In seeking the sparsest classifier, sparse probing may also fail to select important neurons that are redundant within the probing dataset, requiring the use of iterative pruning to enumerate all important neurons. Multi-token features require special processing, often in the form of aggregations that can further weaken the specificity of the result. Finally, depending on how much representations change with scale [79], larger models may utilize more specific features [10], hampering the transferability of datasets between different model scales.

[10] It is also consistent with superposition. Indeed, all of the compound word neurons had higher recall than precision.

6.2 Strengths and Weaknesses of Empirical Findings

In light of the limitations of probing, we attempted to corroborate every case study with independent evidence, including theoretical predictions, ablations, and vocabulary analyses. We find our evidence compelling, although not always conclusive. In particular, we believe we have provided the clearest evidence to date of superposition, monosemantic neurons, and polysemantic neurons in full scale language models. Furthermore, by demonstrating this behavior in seven different models spanning two orders of magnitude in size while exploring over a hundred features, we believe our basic insights are likely to be general and to transfer to current frontier models like GPT-4.

However, much of our analysis is *ad hoc*, tailored to the specific feature being investigated, and requires substantial researcher effort to draw conclusions. While we explored models of varying size, they were all from the same model family and trained with the same data. We think it is unlikely our results are specific to the implementation details of the Pythia model suite, but we do not rule this out. Additionally, the largest model we studied is 6.9 billion parameters which is still more than order-of-magnitude off the frontier. Given the emergent abilities of LLMs with scale [81], it is possible our analysis misses a key dynamic underlying the success of the largest models. Moreover, our results are restricted to binary features and categorical features converted to binary features, and we are not confident that our insights will cleanly transfer to continuous features or cleverly encoded categorical features.

6.3 Implications

For interpretability researchers, our results support the conclusion that superposition is important for the success of models. Hence, attempts to remove it [10] are likely either hiding it or are unlikely to be competitive, but that the highest leverage interventions are perhaps best aimed at the early layers. For AI ethicists and legal scholars, our study of factual neurons point to an important avenue of further inquiry—understanding how neurons encoding protected attributes compute this feature and affect downstream predictions of the model. For AI alignment scholars, our work highlights the potential of identifying safety-critical features and perhaps even manually intervening in computations to enhance our ability to steer models. While none of the features we probed for were especially critical, the Pile subset context neurons could be considered a minimal precursor for situational awareness [47, 82], as these could be used to detect whether an input is from the training or test distribution.

6.4 Future Directions

We only scratched the surface of possible applications and experiments involving sparse probing. Scientifically, further understanding superposition—and how to cope with it—seems central to making progress on the ambitious version of mechanistic interpretability. As outlined in [4], this requires either designing architectures which do not use superposition or to take features out of superposition using a sparse coding like technique—both of which can be assisted with automatic sparse probing as a form of validation. While not explored here, it would also be possible to apply sparse probing to predicting properties of the *output* (e.g. `next_token_is_verb`), as opposed to properties of the input, to better understand the neurons most implicated in making specific types of predictions. With neurons identified in each experiment, it would be potentially insightful to also track how these neurons develop and change through time [83] as well as more carefully analyze how the set of neurons change with scale [79], which is naturally enabled by the Pythia suite's [70] checkpoints. In particular, sparse probing (or similar) is well suited to study neuron splitting in more detail by training probes to predict the value of a less granular neuron from the sum of a sparse set of more granular neurons.

Our primary motivation however, is in ensuring the development of safe AI systems. To further this goal, we envision the development of a large library of probing datasets—potentially with AI assistance—that capture features of particular relevance to bias, fairness, safety, and high-stakes decision making. In addition to automating evaluations of new models, having large and diverse supervised datasets will enable better evaluations of the next generation of unsupervised interpretability techniques [53, 84] that will be needed to keep pace with AI progress.

7 Conclusion

Guided by sparse probing, we have found some of the cleanest examples of monosemanticty, polysemanticity, and superposition in language models "in the wild," and contributed practical guidance and conceptual clarification for how to interpret neurons in greater detail. More than any specific technical contribution, we hope to contribute to the general sense that ambitious interpretability *is* possible—that LLMs have a tremendous amount of rich structure that can and should be understood by humans. We believe this is most productively accomplished with an empirical approach more reminiscent of the natural sciences, such as biology or neuroscience, than the traditional experimental loop of ML. While such research can be hard to finish, it is easy to start, so we encourage the curious researcher to just start looking!